PARISHRAM 2.0 2026 (Class 12th)

Maths

DPP: 2

Inverse Trigonometric Functions

Q1 $\cos^{-1}\frac{4}{5}-\cos^{-1}\frac{15}{17}=$

(A) $\cot^{-1}\frac{84}{85}$

(B) $\tan^{-1}\frac{84}{85}$

(C) $\sin^{-1}\frac{84}{85}$

(D) $\cos^{-1}\frac{84}{85}$

Q2 $\cos^{-1}\frac{12}{13}+\sin^{-1}\frac{3}{5}=$

(A) $\tan^{-1}\frac{65}{56}$

(B) $\sin^{-1}\frac{56}{65}$

(C) $\cos^{-1}\frac{56}{65}$

(D) None of these

Q3 $\cot^{-1}7+\cot^{-1}8+\cot^{-1}18=$

(A) $\cot^{-1}\frac{1}{3}$ (B) $\tan^{-1}3$

(C) $\cot^{-1}3$ (D) None of these

Q4 $\sin\left(2\sin^{-1}0.6\right)=$

(A) 0.81 (B) 0.6

(C) 0.96 (D) 1

Q5 $\tan\left\{2\tan^{-1}\frac{1}{5}-\frac{\pi}{4}\right\}=$

(A) $\frac{7}{17}$ (B) $-\frac{7}{17}$

(C) $\frac{17}{7}$ (D) $-\frac{17}{7}$

Q6 The value of $\cos\left(2\sin^{-1}\frac{3}{5}\right)$ is equal to

(A) $\frac{7}{25}$ (B) $\frac{9}{25}$

(C) $\frac{16}{25}$ (D) 1

Q7 $\tan\left(2\tan^{-1}\frac{1}{5}\right)=$

(A) $\frac{24}{5}$ (B) $\frac{5}{24}$

(C) $\frac{12}{5}$ (D) $\frac{5}{12}$

Q8 $2\sin^{-1}\frac{3}{5}-\tan^{-1}\frac{17}{31}=$

(A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$ (D) $-\frac{\pi}{3}$

Answer Key

Q1	D
Q2	B
Q3	C
Q4	C
Q5	B
Q6	A
Q7	D
Q8	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

$\cos^{-1}\frac{4}{5}-\cos^{-1}\frac{15}{17}$

$=\cos^{-1}\left\{\frac{4}{5}\times\frac{15}{17}+\sqrt{1-\left(\frac{4}{5}\right)^2}\times\sqrt{1-\left(\frac{15}{17}\right)^2}\right\}$

$=\cos^{-1}\left\{\frac{4}{5}\times\frac{15}{17}+\frac{3}{5}\times\frac{8}{17}\right\}$

$=\cos^{-1}\left\{\frac{60}{85}+\frac{24}{85}\right\}=\cos^{-1}\frac{84}{85}$

Video Solution:

Q2 **Text Solution:**

$\cos^{-1}\frac{12}{13}+\sin^{-1}\frac{3}{5}$

$=\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{3}{5}$

$\left[\because \cos^{-1}\frac{12}{13}=\sin^{-1}\frac{5}{13}\right]$

$=\sin^{-1}\left\{\frac{5}{13}\times\sqrt{1-\left(\frac{3}{5}\right)^2}+\frac{3}{5}\times\sqrt{1-\left(\frac{5}{13}\right)^2}\right\}$

$=\sin^{-1}\left\{\frac{5}{13}\times\frac{4}{5}+\frac{3}{5}\times\frac{12}{13}\right\}=\sin^{-1}\frac{56}{65}$

Video Solution:

Q3 **Text Solution:**

$\text{LHS}=\cot^{-1}7+\cot^{-1}8+\cot^{-1}18$

$=\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}+\tan^{-1}\frac{1}{18}$

$\left[\because \cot^{-1}(x)=\tan^{-1}\frac{1}{x},\ \text{if}\ x>0\right]$

$=\left\{\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}\right\}+\tan^{-1}\frac{1}{18}$

$=\tan^{-1}\left\{\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right\}+\tan^{-1}\frac{1}{18}$

$\left[\because xy=\frac{1}{7}\times\frac{1}{8}<1\right]$

$=\tan^{-1}\frac{3}{11}+\tan^{-1}\frac{1}{18}$

$=\tan^{-1}\left\{\frac{\frac{3}{11}+\frac{1}{18}}{1-\frac{3}{11}\times\frac{1}{18}}\right\}$

$\left[\because xy=\frac{3}{11}\times\frac{1}{18}<1\right]$

$=\tan^{-1}\frac{65}{195}=\tan^{-1}\frac{1}{3}=\cot^{-1}3.$

Video Solution:

Q4 **Text Solution:**

$\sin\left(2\sin^{-1}0.6\right)$

$=\sin\left[\sin^{-1}\left\{2\times0.6\times\sqrt{1-(0.6)^2}\right\}\right]$

$\left[\because 2\sin^{-1}x=\sin^{-1}\left(2x\sqrt{1-x^2}\right)\right]$

$=\sin\left(\sin^{-1}0.96\right)=0.96$

Video Solution:

Q5 **Text Solution:**

$\tan\left\{2\tan^{-1}\frac{1}{5}-\frac{\pi}{4}\right\}$

$=\tan\left\{\tan^{-1}\left(\frac{2\times\frac{1}{5}}{1-\frac{1}{25}}\right)-\tan^{-1}1\right\}$

$\left[\because 2\tan^{-1}x=\tan^{-1}\left(\frac{2x}{1-x^2}\right),\ \text{if}\ |x|<1\right]$

$=\tan\left\{\tan^{-1}\frac{5}{12}-\tan^{-1}1\right\}$

$=\tan\left\{\tan^{-1}\left(\frac{\frac{5}{12}-1}{1+\frac{5}{12}}\right)\right\}$

$=\tan\left\{\tan^{-1}\left(\frac{-7}{17}\right)\right\}=\frac{-7}{17}$

Video Solution:

Q6 Text Solution:

Let $\sin^{-1}\frac{3}{5}=\theta$

$\Rightarrow \sin\theta=\frac{3}{5}$

$\therefore \cos\left(2\sin^{-1}\frac{3}{5}\right)=\cos(2\theta)$

$=1-2\sin^2\theta$

$=1-2\left(\frac{3}{5}\right)^2$

$=1-\frac{18}{25}$

$=\frac{7}{25}$

Video Solution:

Q7 Text Solution:

We have,

$\tan\left(2\tan^{-1}\frac{1}{5}\right)=\tan\left\{\tan^{-1}\left(\frac{2\times\frac{1}{5}}{1-\frac{1}{25}}\right)\right\}$

$=\tan\left(\tan^{-1}\frac{5}{12}\right)=\frac{5}{12}$

Video Solution:

Q8 Text Solution:

$2\sin^{-1}\frac{3}{5}-\tan^{-1}\frac{17}{31}$

$=2\tan^{-1}\frac{3}{4}-\tan^{-1}\frac{17}{31}$

$\left[\because \sin^{-1}\frac{3}{5}=\tan^{-1}\frac{3}{4}\right]$

$=\tan^{-1}\left\{\frac{2\times\frac{3}{4}}{1-\left(\frac{3}{4}\right)^2}\right\}-\tan^{-1}\frac{17}{31}$

$\left[\because 2\tan^{-1}x=\tan^{-1}\frac{2x}{1-x^2}\text{ for }|x|<1\right]$

$=\tan^{-1}\frac{24}{7}-\tan^{-1}\frac{17}{13}$

$=\tan^{-1}\left(\frac{\frac{24}{7}-\frac{17}{31}}{1+\frac{24}{7}\times\frac{17}{31}}\right)=\tan^{-1}1=\frac{\pi}{4}$

Video Solution: